

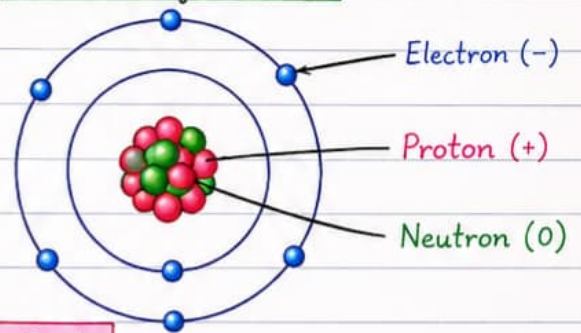
Chapter-12

ATOMS

Introduction

- Atom is the basic unit of matter.
- It is the smallest particle of an element which takes part in chemical reactions.
- Atom consists of **nucleus** (central part) and **electrons** revolving around it.

Structure of an Atom



Key Points

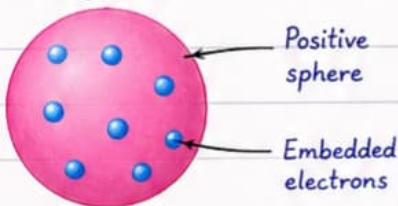
- Number of protons (Z) = Atomic number
- Number of protons + neutrons (A) = Mass number
- For a neutral atom $\rightarrow$ No. of electrons (Z) = No. of protons
- Most of the mass of an atom is concentrated in the nucleus.

Sub-atomic Particles

Particle	Symbol	Charge	Mass (kg)
Electron	e^-	$-1.6 \times 10^{-19} \text{ C}$	9.11×10^{-31}
Proton	p^+	$+1.6 \times 10^{-19} \text{ C}$	1.67×10^{-27}
Neutron	n^0	0	1.67×10^{-27}

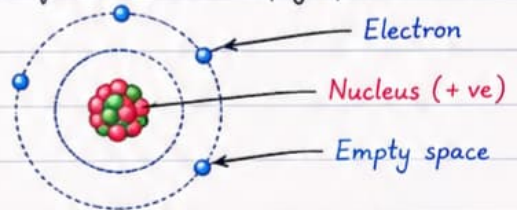
Thomson's Model (Plum Pudding Model)

- Atom is a sphere of positive charge.
- Electrons are embedded in it.
- Overall charge is neutral.



Rutherford's Model (Nuclear Model)

- Atom has a tiny, dense, positively charged nucleus at the center.
- Electrons revolve around the nucleus in large empty space.
- Most of the atom is empty space.



Important Terms

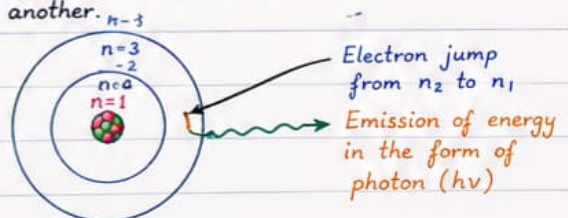
- Atomic number (Z)**: Number of protons in the nucleus.
- Mass number (A)**: Total number of protons and neutrons.
- Isotopes**: Atoms of same element having same Z but different A .
- Isobars**: Atoms of different elements having same A .
- Isotones**: Atoms having same number of neutrons but different Z .

Example

- For carbon atom, $Z = 6$ and $A = 12$.
 - Number of protons = 6
 - Number of neutrons = $A - Z = 12 - 6 = 6$
 - Number of electrons (for neutral atom) = 6

Bohr's Model (Quantised Model)

- Electrons revolve in certain fixed circular orbits called energy levels or shells.
- Each shell has a definite energy.
- No radiation is emitted or absorbed while electron remains in the same shell.
- Radiation is emitted or absorbed only when electron jumps from one shell to another.



★ Atoms are the building blocks of matter. Understanding them helps us understand the world at the smallest scale.

Keep Exploring! ★



Chapter-12 : ATOMS

Tiny in size...
Huge in ideas!



Bohr's Model - Key Postulates

- Electrons revolve around the nucleus in certain stable circular orbits called **stationary states**.
- Only those orbits are allowed for which angular momentum of electron is

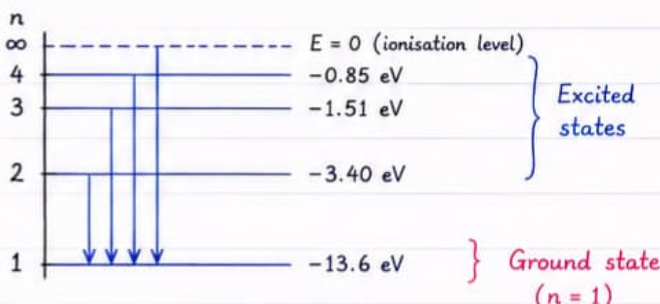
$$mvr = \frac{nh}{2\pi}, \quad n = 1, 2, 3, \dots$$
 where h = Planck's constant
- No radiation is emitted or absorbed as long as electron remains in a stationary orbit.
- Radiation is **emitted** or **absorbed** only when electron jumps from one orbit to another.
- Energy of emitted/absorbed photon is

$$h\nu = |E_i - E_f|$$
 where E_i = initial energy, E_f = final energy

Energy Levels in Bohr's Model

- The energy of electron in n th orbit of a hydrogen atom is

$$E_n = -\frac{13.6 \text{ eV}}{n^2}, \quad n = 1, 2, 3, \dots$$



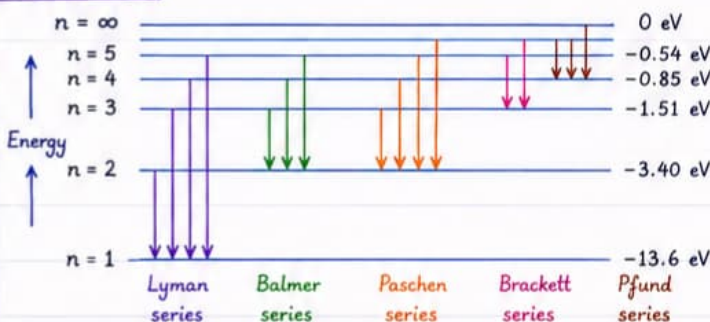
For Hydrogen atom

- ★ $n = 1 \rightarrow$ Ground state (lowest energy)
- ★ $n > 1 \rightarrow$ Excited states (higher energy)

More $n \rightarrow$
Less $|E|$

Spectral Series of Hydrogen Atom

Series	Final level (n_f)	Region	Wavelength Range (approx.)
Lyman series	1	Ultraviolet	91 - 122 nm
Balmer series	2	Visible	400 - 700 nm
Paschen series	3	Infrared	820 nm - 1.9 μm
Brackett series	4	Infrared	1.5 - 4.0 μm
Pfund series	5	Infrared	2.2 - 7.5 μm



★ All these series converge at the ionisation level ($n = \infty$).

de Broglie's Hypothesis

Matter (like electron) shows dual nature.
An electron moving with momentum p has an associated wavelength (λ):

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

where,
 h = Planck's constant ($6.626 \times 10^{-34} \text{ J}\cdot\text{s}$)
 m = mass of electron
 v = velocity of electron

Wave nature helps in understanding atomic stability!

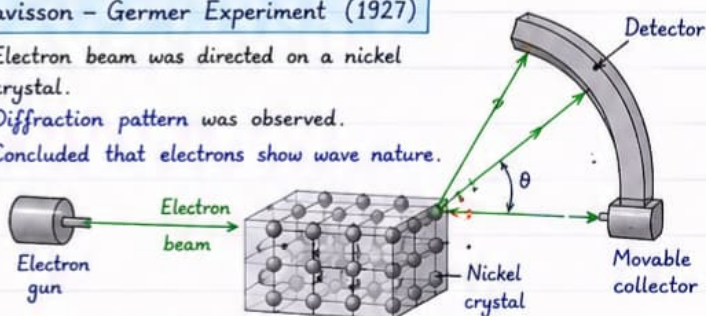
Photoelectric Effect (Brief)

When light of sufficiently high frequency falls on a metal surface, electrons are emitted from the surface.

Explained using quantised energy (photon concept).

Davisson - Germer Experiment (1927)

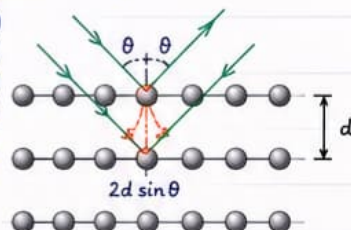
- Electron beam was directed on a nickel crystal.
- Diffraction pattern was observed.
- Concluded that electrons show wave nature.



Bragg's condition for maximum intensity

$$n\lambda = 2d \sin \theta$$

where d = distance between crystal planes.
 $n = 1, 2, 3, \dots$



Key Takeaway

→ Atoms are not indivisible. They have a nuclear centre and electrons revolve around it in quantised orbits. Understanding of atoms evolved with experiments and new ideas!





Chapter-12 : ATOMS

Think Tiny,
Understand Big!



Spectral Lines of Hydrogen Atom

- When electrons jump from higher energy level (n_2) to lower energy level (n_1), spectral lines are produced.

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) ; \quad n_2 > n_1$$

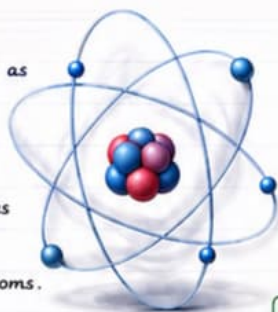
where R = Rydberg constant = $1.097 \times 10^7 \text{ m}^{-1}$

Series	Lower level (n_1)	Region	Colour	Example transition
Lyman	1	Ultraviolet	—	$n_2 \rightarrow 1$
Balmer	2	Visible (VIBGYOR)	—	$n_2 \rightarrow 2$
Paschen	3	Infrared	—	$n_2 \rightarrow 3$
Brackett	4	Infrared	—	$n_2 \rightarrow 4$
Pfund	5	Infrared	—	$n_2 \rightarrow 5$

☆ The wavelengths in a series converge towards a limiting value as $n_2 \rightarrow \infty$.

Drawbacks of Bohr's Model

- Could not explain stability of atoms as per classical physics.
- Failed to explain fine structure of spectral lines.
- Could not explain relative intensities of spectral lines.
- Not applicable to multi-electron atoms.
- Did not account for external fields (Zeeman & Stark effect).



Quantum Numbers

Quantum numbers describe the state of an electron in an atom.

Quantum Number	Symbol	What it describes	Allowed Values
Principal	n	Energy level / Shell	$n = 1, 2, 3, \dots \infty$
Azimuthal (Subsidiary)	l	Shape of orbital (subshell)	$l = 0 \text{ to } (n-1)$ (i.e., $0, 1, \dots, n-1$)
Magnetic	m_l	Orientation of orbital in space	$m_l = -l \text{ to } +l$ (integer values)
Spin	m_s	Spin direction of electron	$m_s = +\frac{1}{2} \text{ or } -\frac{1}{2}$

Electronic Configuration (Brief)

- Distribution of electrons in different shells and subshells.
- Aufbau principle, Pauli exclusion principle and Hund's rule govern the filling.
- Order of filling (first 20 elements):

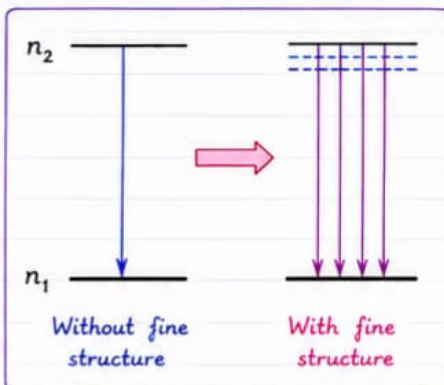
$$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s \dots$$



From tiny atoms to vast universes - curiosity is the strongest energy! ☆

Fine Structure of Spectral Lines

- Spectral lines are not perfectly sharp.
- Each line actually consists of a group of closely spaced lines.
- Due to spin-orbit coupling and other interactions.



Remember!

Energy decreases
↓
Wavelength increases (red shift)
Energy increases
↓
Wavelength decreases (blue shift)

Quantum Mechanical Model (Modern View)

- Electrons do not revolve in fixed orbits.
- They exist in orbitals → regions of maximum probability of finding an electron.
- Based on the uncertainty principle.
- Described by wave function (Ψ).

Heisenberg's Uncertainty Principle

It is impossible to determine simultaneously the exact position and momentum of an electron.

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

where Δx = uncertainty in position

Δp = uncertainty in momentum

h = Planck's constant

Shapes of Some Common Orbitals

s-orbital	p_x -orbital	p_y -orbital	p_z -orbital
Spherical (symmetrical)	Dumb-bell (along x-axis)	Dumb-bell (along y-axis)	Dumb-bell (along z-axis)

Example

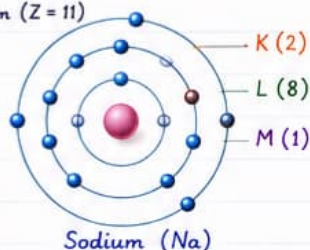
Write electronic configuration of Sodium ($Z = 11$)

Solution:

Fill according to order →

$$1s^2 2s^2 2p^6 3s^1$$

2 8 1
K L M





Chapter-12 : ATOMS



Small scale...
Big Wonder! ♥

X-rays and Atomic Number

- When high energy electrons strike a metal target, X-rays are produced.
- These X-rays are characteristic of the target element.
- Moseley (1913) studied the spectra of X-rays from different elements.
- He found that $\sqrt{\nu}$ is directly proportional to $(Z - b)$.

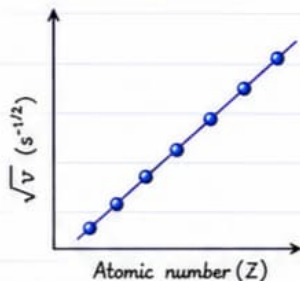
$$\sqrt{\nu} = a(Z - b)$$

where a and b are constants.

Thus, atomic number (Z) is the fundamental property of an element, not its atomic mass.

- ★ Moseley showed that K_{α} line frequency increases regularly with atomic number.

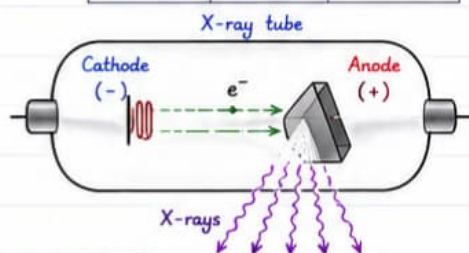
Moseley's Plot



Straight line confirms Z is more fundamental than atomic mass.

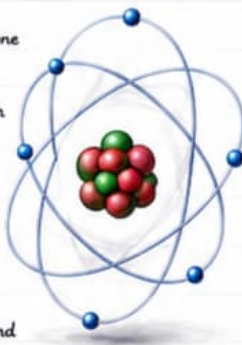
Some Characteristic X-ray Lines

Series	Line	Transition
K-series	K_{α}	$L \rightarrow K$
	K_{β}	$M \rightarrow K$
L-series	L_{α}	$M \rightarrow L$
	L_{β}	$N \rightarrow L$
M-series	M_{α}	$N \rightarrow M$



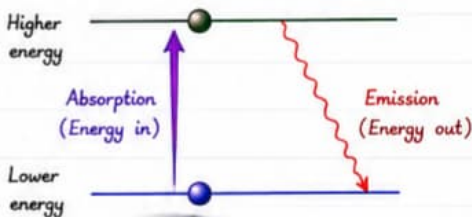
Limitations of Bohr's Model

- Applicable only to hydrogen like atoms (one electron systems).
- Fails to explain the spectra of multielectron atoms.
- Does not account for fine structure of spectral lines.
- Cannot explain the relative intensities of spectral lines.
- Does not incorporate the wave nature and uncertainty principle.



Why Do Atoms Exist?

- Atoms exist because they are in a state of minimum potential energy (most stable state).
- When energy is given to an atom, it absorbs it and jumps to higher energy level.
- When it returns to lower energy level, energy is released in the form of radiation.



Absorb to excite,
Emit to stabilize!

Summary

- ★ Atom is made up of nucleus (protons + neutrons) and electrons revolving around it.
- ★ Electrons revolve in certain fixed energy orbits (Bohr's postulates).
- ★ Energy is emitted or absorbed when an electron jumps between orbits.
- ★ Hydrogen spectrum explained by Bohr's model and gives rise to spectral series.
- ★ Bohr model was improved by introducing fine structure and quantum ideas.
- ★ Modern view: electrons exist in orbitals described by quantum mechanics based on uncertainty principle.

Important Formulae

Formula	Meaning
$mvr = \frac{nh}{2\pi}$	Quantisation of angular momentum
$E_n = -\frac{13.6}{n^2} \text{ eV}$	Energy of electron in n th orbit (H-atom)
$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$	Rydberg formula (spectral lines)
$E = h\nu = \frac{hc}{\lambda}$	Energy of photon
$\sqrt{\nu} = a(Z - b)$	Moseley's equation (X-ray spectra)
$\Delta x \Delta p \geq \frac{h}{4\pi}$	Heisenberg uncertainty principle

Key Points to Remember

- ★ Atomic number (Z) = No. of protons in nucleus.
- ★ Mass number (A) = Protons + Neutrons.
- ★ Isotopes: Same Z , different A .
- ★ Isobars: Same A , different Z .
- ★ Isotones: Same neutrons, different Z .

NEET Corner

High yield chapter!
Don't miss any concept.

Practice Example

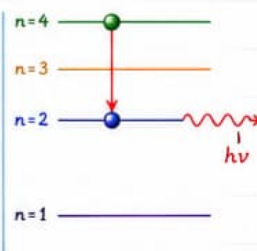
Q. An electron in hydrogen atom drops from $n = 4$ to $n = 2$. Calculate the wavelength of emitted radiation.

Solution:

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = R \left(\frac{4-1}{16} \right) = \frac{3R}{16}$$

$$\lambda = \frac{16}{3R} = \frac{16}{3 \times 1.097 \times 10^7} = 4.86 \times 10^{-7} \text{ m}$$

486 nm (in visible region)



Understanding atoms is understanding the building blocks of everything around us.

Keep Exploring, Keep Questioning! 😊

